

Tutorial Set I

Department of Statistics and Actuarial Science
STATS 339

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SECTION A

Use Elementary Row Operations to solve the following systems of equations

1) $x + y + 2z = 4$
 $x - 2y + z = 0$
 $x - 5y = -4$

2) $x + y + z = 4$
 $x + 2y + 4z = 12$
 $2x - 3y - z = 4$

3) $x + 2y + 3z = 14$
 $3x + 2y + z = 10$
 $3x + y + 2z = 11$

4) $a - 3b - 26c + 22d = 0$
 $a - 8c + 7d = 0$
 $a + b - 2c + 2d = 0$
 $4a + 5b - 2c + 3d = 0$

5) $2a + 5b + 4c + d = 20$
 $a + 3b + 2c + d = 11$
 $2a + 10b + 9c + 7d = 40$
 $3a + 8b + 9c + 2d = 37$

6) $a + 2b + 3c + 4d = 0$
 $7a + 14b + 20c + 27d = 0$
 $5a + 10b + 16c + 19d = -2$
 $3a + 5b + 6c + 13d = 5$

SECTION B

Use Elementary Row Operations to find the inverse of the following matrices

1) $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

2) $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

$$3) \ A = \begin{bmatrix} 1 & 1 & 1 \\ 6 & 5 & 4 \\ 13 & 10 & 8 \end{bmatrix}$$

$$4) \ A = \begin{bmatrix} 2 & 5 & 7 \\ 6 & 3 & 4 \\ 5 & -2 & -3 \end{bmatrix}$$

SECTION C

Solve the equation given by the determinant.

$$1) \ \begin{vmatrix} x-1 & 4 \\ 1 & x+2 \end{vmatrix} = 0$$

$$2) \ \begin{vmatrix} x^2+1 & x \\ x(x-3) & x-2 \end{vmatrix} = 0$$

$$3) \ \begin{vmatrix} x & 1 & x+1 \\ 2 & x & 3 \\ x+1 & 4 & x \end{vmatrix} = -2x^2 + 11$$

$$4) \ \begin{vmatrix} 4-x & 1 & 0 \\ 2 & 6-x & 1 \\ 0 & 1 & 4-x \end{vmatrix} = 0$$

Section D

Find the sum, product and difference of the following matrices

$$1) \ A = \begin{pmatrix} 43 & -62 & 76 \\ 54 & -10 & 57 \\ 90 & -9 & 7 \end{pmatrix}, \ B = \begin{pmatrix} 31 & -7 & -2 \\ -6 & 8 & 2 \\ -28 & 49 & 99 \end{pmatrix}$$

$$2) \ A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & -1 & -2 & 3 \\ 4 & -3 & 1 & 5 \\ 2 & -4 & -1 & 0 \end{pmatrix}, \ B = \begin{pmatrix} 5 & 3 & -1 & 8 \\ 2 & 0 & 7 & 4 \\ -2 & -3 & -3 & -5 \\ 4 & 1 & 1 & -2 \end{pmatrix}$$

$$3) \ A = \begin{pmatrix} 3 & 8 & -2 \\ 4 & 1 & 0 \\ 2 & 3 & 8 \end{pmatrix}, \ B = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$$4) \ A = \begin{pmatrix} 3 & 4 & 1 \\ 0 & -2 & 3 \\ 2 & 1 & -1 \end{pmatrix}, \ B = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 3 & 8 \\ 4 & 1 & -2 \end{pmatrix}$$

$$5) \ A = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 2 & 4 \\ 2 & 1 & 1 \end{pmatrix}, \ B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 3 & 2 \\ 2 & 0 & 1 \end{pmatrix}$$

$$6) \ A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 3 & 1 \end{pmatrix}, \ B = \begin{pmatrix} 1 & 3 \\ 1 & 4 \\ 2 & 2 \end{pmatrix}$$

SECTION E

Find the Determinant, Inverse and Reduced Row Echelon Form of the following matrices.

$$1) \ \begin{pmatrix} 5 & 1 & -2 \\ -2 & 0 & 5 \\ 7 & 2 & 8 \end{pmatrix}$$

$$2) \ \begin{pmatrix} 0 & 5 & 5 \\ 1 & 6 & 5 \\ -2 & 8 & 9 \end{pmatrix}$$

$$3) \ \begin{pmatrix} 10 & 20 & 30 \\ 7 & 18 & 20 \\ -2 & -7 & -5 \end{pmatrix}$$

$$4) \ \begin{pmatrix} 5 & -5 & 5 & 6 \\ 2 & 1 & 1 & 2 \\ -1 & -1 & 0 & 1 \\ 5 & 1 & 2 & 1 \end{pmatrix}$$

$$5) \ \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 \end{pmatrix}$$

$$6) \ \begin{pmatrix} 3 & 2 & 1 & 2 \\ 7 & 5 & 2 & 5 \\ 0 & 0 & 9 & 4 \\ 0 & 0 & 11 & 5 \end{pmatrix}$$